

Tet3 Deletion in Adult Brain Neurons of Female Mice Results in Anxiety-like Behavior and Cognitive Impairments.

PMID: 35665901 | Indexed in PubMed

TET enzymes (TET1-3) are dioxygenases that oxidize 5-methylcytosine (5mC) into 5-hydroxymethylcytosine (5hmC) and are involved in the DNA demethylation process. In line with the observed 5hmC abundance in the brain, Tet genes are highly transcribed, with Tet3 being the predominant member. We have previously shown that Tet3 conditional deletion in the brain of male mice was associated with anxiety-like behavior and impairment in hippocampal-dependent spatial orientation. In the current study, we addressed the role of Tet3 in female mice and its impact on behavior, using in vivo conditional and inducible deletion from post-mitotic neurons. Our results indicate that conditional and inducible deletion of Tet3 in female mice increases anxiety-like behavior and impairs both spatial orientation and short-term memory. At the molecular level, we identified upregulation of immediate-early genes, particularly Npas4, in both the dorsal and ventral hippocampus and in the prefrontal cortex. This study shows that deletion of Tet3 in female mice differentially affects behavioral dimensions as opposed to Tet3 deletion in males, highlighting the importance of studying both sexes in behavioral studies. Moreover, it contributes to expand the knowledge on the role of epigenetic regulators in brain function and behavioral outcome.

Molecular basis of an atypical dsDNA 5mC/6mA bifunctional dioxygenase CcTet from Coprinopsis cinerea in catalyzing dsDNA 5mC demethylation.

PMID: 38324471 | Indexed in PubMed

The eukaryotic epigenetic modifications 5-methyldeoxycytosine (5mC) and N6-methyldeoxyadenine (6mA) have indispensable regulatory roles in gene expression and embryonic development. We recently identified an atypical bifunctional dioxygenase CcTet from *Coprinopsis cinerea* that works on both 5mC and 6mA demethylation. The nonconserved residues Gly331 and Asp337 of CcTet facilitate 6mA accommodation, while D337F unexpectedly abolishes 5mC oxidation activity without interfering 6mA demethylation, indicating a prominent distinct but unclear 5mC oxidation mechanism to the conventional Tet enzymes. Here, we assessed the molecular mechanism of CcTet in catalyzing 5mC oxidation by representing the crystal structure of CcTet-5mC-dsDNA complex. We identified the distinct mechanism by which CcTet recognizes 5mC-dsDNA compared to 6mA-dsDNA substrate. Moreover, Asp337 was found to have a central role in compensating for the loss of a critical 5mC-stabilizing H-bond observed in conventional Tet enzymes, and stabilizes 5mC and subsequent intermediates through an H-bond with the N4 atom of the substrates. These findings improve our understanding of Tet enzyme functions in the dsDNA 5mC and 6mA demethylation pathways, and provide useful information for future discovery of small molecular probes targeting Tet enzymes in DNA active demethylation processes.

Inducible disruption of Tet genes results in myeloid malignancy, readthrough transcription, and a heterochromatin-to-euchromatin switch.

PMID: 37406303 | Indexed in PubMed

The three mammalian TET dioxygenases oxidize the methyl group of 5-methylcytosine in DNA, and the

oxidized methylcytosines are essential intermediates in all known pathways of DNA demethylation. To define the in vivo consequences of complete TET deficiency, we inducibly deleted all three Tet genes in the mouse genome. Tet1/2/3-inducible TKO (iTKO) mice succumbed to acute myeloid leukemia (AML) by 4 to 5 wk. Single-cell RNA sequencing of Tet iTKO bone marrow cells revealed the appearance of new myeloid cell populations characterized by a striking increase in expression of all members of the stefin/cystatin gene cluster on mouse chromosome 16. In patients with AML, high stefin/cystatin gene expression correlates with poor clinical outcomes. Increased expression of the clustered stefin/cystatin genes was associated with a heterochromatin-to-euchromatin compartment switch with readthrough transcription downstream of the clustered stefin/cystatin genes as well as other highly expressed genes, but only minor changes in DNA methylation. Our data highlight roles for TET enzymes that are distinct from their established function in DNA demethylation and instead involve increased transcriptional readthrough and changes in three-dimensional genome organization.

Role of DNA De-methylation intermediate '5-hydroxymethylcytosine' in ovarian cancer management: A comprehensive review.

PMID: 36099791 | Indexed in PubMed

Ovarian cancer remains the most eminent silent killer, with high morbidity and mortality among all gynaecological cancers. The advanced-stage patient's diagnosis has a low survival rate caused by its asymptomatic progression and diverse histopathological sub-types, wherefore in poor prognosis and highly recurring malignancy with multidrug resistance towards chemotherapy. Epigenetic biomarkers open promising avenues of intriguing research to combat OC malignancy, furthermore a tool for its early diagnosis. 5-hydroxymethylcytosine (5-hmC), alias the sixth base of the genome, is an intermediate formed during the recently established DNA demethylation process and catalysed via ten-eleven translocation (TET) family of enzymes. It plays a significant role in regulating gene expression and has sparked interest in various cancer types. This review summarizes the role of active DNA demethylation process, its enzymes and intermediate 5-hmC in epigenetic landscape of ovarian cancer as a potent biomarker for clinical translation in identification of therapeutic targets, diagnostic and prognostic evaluation.

Pronounced sequence specificity of the TET enzyme catalytic domain guides its cellular function.

PMID: 36070377 | Indexed in PubMed

TET (ten-eleven translocation) enzymes catalyze the oxidation of 5-methylcytosine bases in DNA, thus driving active and passive DNA demethylation. Here, we report that the catalytic domain of mammalian TET enzymes favor CGs embedded within basic helix-loop-helix and basic leucine zipper domain transcription factor-binding sites, with up to 250-fold preference in vitro. Crystal structures and molecular dynamics calculations show that sequence preference is caused by intrasubstrate interactions and CG flanking sequence indirectly affecting enzyme conformation. TET sequence preferences are physiologically relevant as they explain the rates of DNA demethylation in TET-rescue experiments in culture and in vivo within the zygote and germ line. Most and least favorable TET motifs represent DNA sites that are bound by methylation-sensitive immediate-early transcription factors and octamer-binding

transcription factor 4 (OCT4), respectively, illuminating TET function in transcriptional responses and pluripotency support.

TET enzyme driven epigenetic reprogramming in early embryos and its implication on long-term health.

PMID: 39149518 | Indexed in PubMed

Mammalian embryo development is initiated by the union of paternal and maternal gametes. Upon fertilization, their epigenome landscape is transformed through a series of finely orchestrated mechanisms that are crucial for survival and successful embryogenesis. Specifically, maternal or oocyte-specific reprogramming factors modulate germ cell specific epigenetic marks into their embryonic states. Rapid and dynamic changes in epigenetic marks such as DNA methylation and histone modifications are observed during early embryo development. These changes govern the structure of embryonic genome prior to zygotic genome activation. Differential changes in epigenetic marks are observed between paternal and maternal genomes because the structure of the parental genomes allows interaction with specific oocyte reprogramming factors. For instance, the paternal genome is targeted by the TET family of enzymes which oxidize the 5-methylcytosine (5mC) epigenetic mark into 5-hydroxymethylcytosine (5hmC) to lower the level of DNA methylation. The maternal genome is mainly protected from TET3-mediated oxidation by the maternal factor, STELLA. The TET3-mediated DNA demethylation occurs at the global level and is clearly observed in many mammalian species. Other epigenetic modulating enzymes, such as DNA methyltransferases, provide fine tuning of the DNA methylation level by initiating de novo methylation. The mechanisms which initiate the epigenetic reprogramming of gametes are critical for proper activation of embryonic genome and subsequent establishment of pluripotency and normal development. Clinical cases or diseases linked to mutations in reprogramming modulators exist, emphasizing the need to understand mechanistic actions of these modulators. In addition, embryos generated via in vitro embryo production system often present epigenetic abnormalities. Understanding mechanistic actions of the epigenetic modulators will potentially improve the well-being of individuals suffering from these epigenetic disorders and correct epigenetic abnormalities in embryos produced in vitro. This review will summarize the current understanding of epigenetic reprogramming by TET enzymes during early embryogenesis and highlight its clinical relevance and potential implication for assisted reproductive technologies.

Target specificity of mammalian DNA methylation and demethylation machinery.

PMID: 29265160 | Indexed in PubMed

DNA methylation is an essential epigenetic modification for mammalian embryonic development and biology. The DNA methylation pattern across the genome, together with other epigenetic signals, is responsible for the transcriptional profile of a cell and thus preservation of the cell's identity. Equally, the family of TET enzymes which triggers the initiation of the DNA demethylation cycle plays a vital role in the early embryonic development and a lack of these enzymes at later stages leads to a diseased state and dysregulation of the epigenome. DNA methylation has long been considered a very stable modification; however, it has become increasingly clear that for the establishment and maintenance of the methylation pattern, both generation of DNA methylation and its removal are important, and that a

delicate balance of ongoing DNA methylation and demethylation shapes the final epigenetic methylation pattern of the cell. Although this epigenetic mark has been investigated in great detail, it still remains to be fully understood how specific DNA methylation imprints are precisely generated, maintained, read or erased in the genome. Here, we provide a biochemist's view on how both DNA methyltransferases and TET enzymes are recruited to specific genomic loci, and how their chromatin interactions, as well as their intrinsic sequence specificities and molecular mechanisms, contribute to the methylation pattern of the cell.

The Role of DNMT Methyltransferases and TET Dioxygenases in the Maintenance of the DNA Methylation Level.

PMID: 39334883 | Indexed in PubMed

This review deals with the functional characteristics and biological roles of enzymes participating in DNA methylation and demethylation as key factors in epigenetic regulation of gene expression. The set of enzymes that carry out such processes in human cells is limited to representatives of two families, namely DNMT (DNA methyltransferases) and TET (DNA dioxygenases). The review presents detailed information known today about each functionally important member of these families and describes the catalytic activity and roles in the mammalian body while also providing examples of dysregulation of the expression and/or activity of these enzymes in conjunction with the development of some human disorders, including cancers, neurodegenerative diseases, and developmental pathologies. By combining the up-to-date information on the dysfunction of various enzymes that control the DNA "methylome" in the human body, we hope not only to draw attention to the importance of the maintenance of a required DNA methylation level (ensuring epigenetic regulation of gene expression and normal functioning of the entire body) but also to help identify new targets for directed control over the activity of the enzymes that implement the balance between processes of DNA methylation and demethylation.

[Oxidation and deamination of nucleobases as an epigenetic tool].

PMID: 22706113 | Indexed in PubMed

Recent discoveries have demonstrated that 5-methylcytosine (5mC) may be hydroxymethylated to 5-hydroxymethylcytosine (5hmC) in mammals and that genomic DNA may contain about 0.02-0.7% of 5hmC. The aforementioned modification is the key intermediate of active DNA demethylation and has been named "the sixth base in DNA". Although active DNA demethylation in mammals is still controversial, the most plausible mechanism/s of active 5mC demethylation include involvement of three families of enzymes; i) Tet, which is involved in hydroxylation of 5mC to form 5hmC, which can be further oxidized to 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC); ii) deamination of 5mC (or 5hmC) by AID/APOBEC to form thymine or 5-hydroxymethyluracil (5hmU) mispaired with guanine; iii) the BER pathway induced by involvement of TDG glycosylase to replace the above described base modification (5fC, 5caC, 5hmU) with cytosine to demethylate DNA. A plausible scenario for engagement of TDG glycosylase (or some other G-T glycosylase) is through prior deamination of 5-mC to thymine, which generates a G: T substrate for the enzyme. Here cytidine deaminase of the AID/APOBEC family was implicated in the deamination step. It is possible that TDG may act in concert with these deaminases. It seems that mutations are not the only effect of oxidatively modified DNA bases. These, as yet,

understudied aspects of the damage suggest a potential for 8-oxoguanine (8-oxoGua) to affect gene expression via chromatin relaxation. It is possible that 8-oxoGua presence in specific DNA sequences may be widely used for transcription regulation, which suggests the epigenetic nature of 8-oxoGua presence in DNA.

Analysis of Methylation Dynamics Reveals a Tissue-Specific, Age-Dependent Decline in 5-Methylcytosine Within the Genome of the Vertebrate Aging Model *Nothobranchius furzeri*.

PMID: 34222326 | Indexed in PubMed

Erosion of the epigenetic DNA methylation landscape is a widely recognized hallmark of aging. Emerging advances in high throughput sequencing techniques, in particular DNA methylation data analysis, have resulted in the establishment of precise human and murine age prediction tools. In vertebrates, methylation of cytosine at the C5 position of CpG dinucleotides is executed by DNA methyltransferases (DNMTs) whereas the process of enzymatic demethylation is highly dependent on the activity of the ten-eleven translocation methylcytosine dioxygenase (TET) family of enzymes. Here, we report the identification of the key players constituting the DNA methylation machinery in the short-lived teleost aging model *Nothobranchius furzeri*. We present a comprehensive spatio-temporal expression profile of the methylation-associated enzymes from embryogenesis into late adulthood, thereby covering the complete killifish life cycle. Data mining of the *N. furzeri* genome produced five *dnmt* gene family orthologues corresponding to the mammalian DNMTs (DNMT1, 2, 3A, and 3B). Comparable to other teleost species, *N. furzeri* harbors multiple genomic copies of the de novo DNA methylation subfamily. A related search for the DNMT1 recruitment factor UHRF1 and TET family members resulted in the identification of *N. furzeri* *uhrf1*, *tet1*, *tet2*, and *tet3*. Phylogenetic analysis revealed high cross-species similarity on the amino acid level of all individual *dnmts*, *tets*, and *uhrf1*, emphasizing a high degree of functional conservation. During early killifish development all analyzed *dnmts* and *tets* showed a similar expression profile characterized by a strong increase in transcript levels after fertilization, peaking either at embryonic day 6 or at the black eye stage of embryonic development. In adult *N. furzeri*, DNA methylation regulating enzymes showed a ubiquitous tissue distribution. Specifically, we observed an age-dependent downregulation of *dnmts*, and to some extent *uhrf1*, which correlated with a significant decrease in global DNA methylation levels in the aging killifish liver and muscle. The age-dependent DNA methylation profile and spatio-temporal expression characteristics of its enzymatic machinery reported here may serve as an essential platform for the identification of an epigenetic aging clock in the new vertebrate model system *N. furzeri*.